

EduGenie Solution Requirements

Functional Requirements

ID	Requirement
FR-01	The user can open the EduGenie homepage.
FR-02	The user can ask an educational question.
FR-03	The system can explain a concept according to the learner's level.
FR-04	The system can generate a topic-specific quiz.
FR-05	The system can summarize educational text.
FR-06	The system can create a personalized learning roadmap.
FR-07	The system validates user input.
FR-08	The system displays useful error messages.
FR-09	The system uses Google Gemini for AI generation.
FR-10	The system displays results in the frontend.

Non-Functional Requirements

ID	Requirement
NFR-01	The interface should be responsive on desktop and mobile screens.
NFR-02	API keys must not be stored in frontend files.
NFR-03	The application should use modular Python files.
NFR-04	The backend should return appropriate HTTP status codes.
NFR-05	User input should have length limits.
NFR-06	The application should provide clear error messages.
NFR-07	The application should be easy to maintain.
NFR-08	The application should support future database integration.
NFR-09	The application should run on a normal development laptop.
NFR-10	The application should be documented and tested.

API Requirements

Method	Endpoint	Purpose
GET	/	Serve the frontend

Method	Endpoint	Purpose
GET	/api/health	Check application health
POST	/api/ask	Answer a question
POST	/api/explain	Explain a concept
POST	/api/quiz	Generate a quiz
POST	/api/summarize	Summarize text
POST	/api/roadmap	Create a learning roadmap

Validation Requirements

- Questions must contain at least three characters.
- Concepts must contain at least two characters.
- Summary text must contain at least fifty characters.
- Quiz question count must be between one and fifteen.
- Weekly study hours must be between one and sixty.